

8.5 FOOD CHAINS, FOOD WEB, PRODUCTIVITY AND ENERGY FLOW

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An ecosystem consists of structural and functional components. The structural components are **abiotic** and **biotic** factors. You have studied various abiotic components like light, temperature, water, pressure etc., of ecosystem in the previous lessons. The biotic components of an ecosystem are composed of producers, consumers, and decomposers. The functional components of ecosystem are **cycling of nutrients** and the **flow of energy**. Let us look at these components in a more integrated manner and see how the flow of energy takes place within these components of the ecosystem.

8.5.1 Biotic Factors

1. Producers

The green plants in the ecosystem-terminology are called **producers**. They synthesize the food by using solar energy (photo autotrophs). Certain bacteria such as Iron and Sulphur bacteria are **chemoautotrophs**. They fix carbon dioxide by using the energy obtained by the breaking down of certain chemical substances. In a terrestrial ecosystem, the major producers are the herbaceous and woody plants. Likewise, primary producers in an aquatic ecosystem include various species like microscopic phytoplanktonic organisms, algae and some higher plants. The producers form the first trophic level in most food chains. They constitute the main source of food for the organisms of the second/ next higher trophic level, the 'consumers'.

2. Consumers

All animals depend on plants (directly or indirectly) for their food needs. They are hence called consumers and also **heterotrophs**. If they feed on the producers - (the plants) they are called **primary consumers**. If the animals eat other animals, which eat plants or plant products, they are called **secondary consumers**. Likewise, there may be **tertiary consumers** too. Obviously the primary consumers are the herbivores. Some common herbivores are insects, birds and mammals in the terrestrial ecosystem and arthropods and molluscs in the aquatic ecosystems.

3. Decomposers

You may have heard of the earthworm being referred to as the farmer's 'friend'. This is so because they help in the breakdown of complex organic matter as well as in loosening of the soil (making the soil porous). Similarly, decomposers break down complex organic matter into simple inorganic substances like carbon dioxide, water and nutrients and the process is called **decomposition**. Plant parts such as leaves, bark, flowers and dead remains of animals, including their faecal matter, constitute the **detritus**. Detritus is the '**raw material**' for '**decomposition**'.

The important steps in the process of decomposition are **fragmentation**, **leaching**, **catabolism**, **humification** and **mineralization**.

Detritivores (e.g. earthworm) break down detritus (litter) into smaller particles and this process is called **fragmentation**. By the process of **leaching**, water soluble inorganic nutrients go down into the soil and get precipitated as unavailable salts. Bacterial and fungal enzymes degrade detritus into simpler inorganic substances. This process is called **catabolism**.

It is important to note that all the above steps in decomposition operate simultaneously on the detritus. Humification and mineralization occur during decomposition in the soil. Humification leads to accumulation of a dark coloured amorphous substance called **humus** that is highly resistant to microbial action and undergoes decomposition at an extremely slow rate. Being colloidal in nature it serves as a **reservoir of nutrients**. The humus is further degraded by some microbes and release of inorganic nutrients occurs by the process known as **mineralization**. (Ref: NCERT Text Book pages 243-244)

Decomposition

Decomposition is largely an oxygen-requiring process. The rate of decomposition is controlled by chemical composition of the detritus and *climatic factors*. In a particular climatic condition, decomposition rate is slower if detritus is rich in **lignin**.

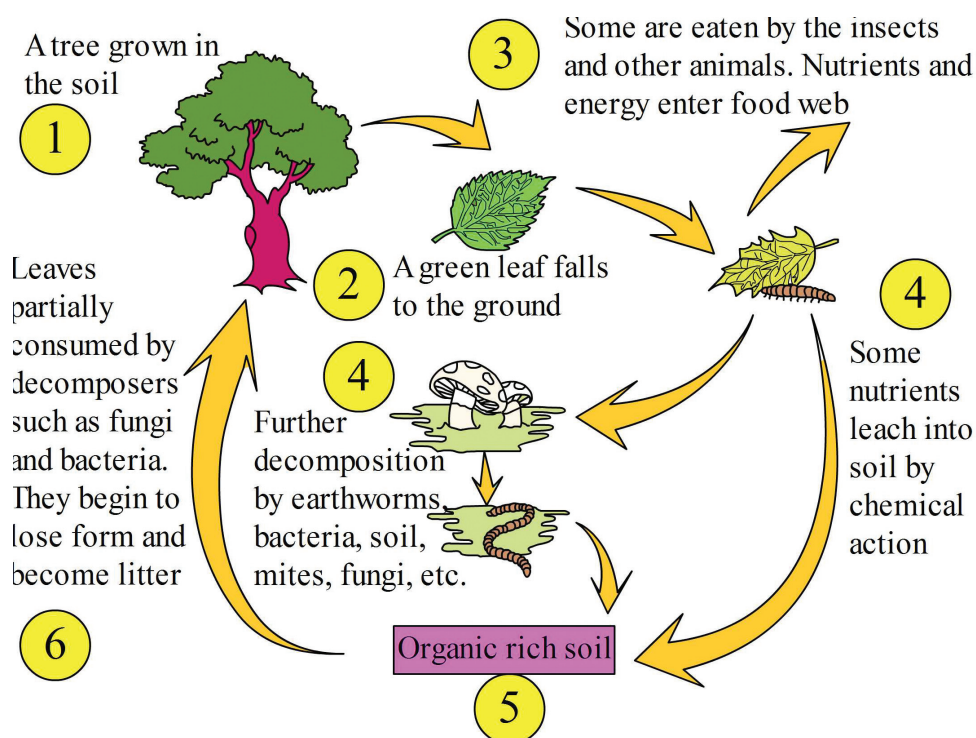
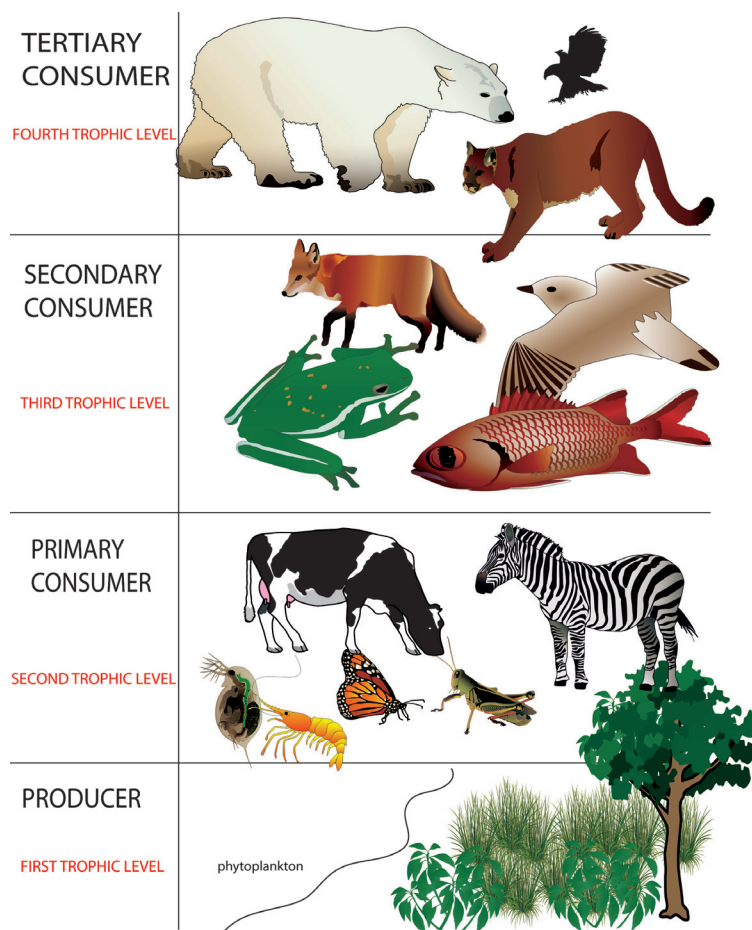


Fig. 8.12 Diagrammatic representation of decomposition cycle in a terrestrial ecosystem (NCERT)

and **chitin**, and quicker, if detritus is rich in **nitrogen** and water-soluble substances like **sugars**. Temperature and soil moisture are the most important climatic factors that regulate decomposition through their effects on the activities of soil microbes. Warm and moist environment favours decomposition whereas **low temperature** and **anaerobic environment** 'inhibit' the decomposition resulting in build up of organic materials. As most of the decomposers are very small microscopic forms, they are also called 'micro-consumers'.

8.5.2 The Food Chains

Energy flows into biological systems (ecosystems) from the Sun. The biological systems of environment include several food levels called **trophic levels**. A trophic level is composed of those organisms which have the same source of energy and having the same number of steps (energy storing and transfer steps) away from the sun. Thus a plant's trophic level is **one**, while that of a herbivore - **two**, and that of the first level carnivore - **three**. The second and third levels of the carnivores occupy **fourth** and **fifth** trophic levels respectively.



A given organism may occupy more than one trophic level simultaneously because of its food habits. One must remember that the trophic level represents a functional level. A given species may occupy more than one trophic level in the same ecosystem at the same time; for example, a sparrow is a primary consumer when it eats seeds, fruits, and a secondary consumer when it eats insects and worms (omnivore).

The food energy passes from one trophic level to another trophic level mostly from the lower to higher trophic levels. When the path of food energy is 'linear', the components resemble the 'links' of a chain, and it is called '**food chain**'. Generally a food chain ends with decomposers. The three major types of food chains in an ecosystem are Grazing Food Chain, Parasitic Food Chain and Detritus Food Chain.

Fig. 8.13 Trophic levels in an ecosystem

I. Grazing Food Chain

It is also known as predatory food chain. It begins with the green plants (producers) and the second, third and fourth trophic levels are occupied by the herbivores, primary carnivores and secondary carnivores respectively. In some food chains there is yet another trophic level – the climax carnivores. The number of trophic levels in food chains varies from 3 to 5 generally. Some examples for grazing food chain (**GFC**) are given below.

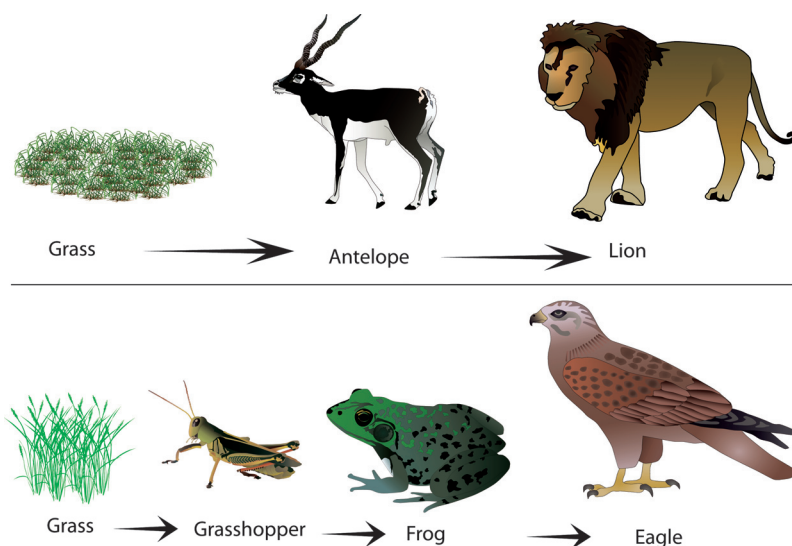


Fig. 8.14 Grazing food chain

I trophic level	II trophic level	III trophic level	IV trophic level	V trophic level
Rosebush	→ Aphids	→ Spiders	→ Small birds	→ Hawk
Grass	→ Grasshopper	→ Frog	→ Snake	→ Hawk
Plants	→ Caterpillar	→ Lizard	→ Snake	
Phytoplankton	→ Zooplankton	→ Fish	→ Bird	
Grass	→ Goat	→ Man		

II. Parasitic food chain

Some authors included the 'Parasitic Food Chains' as a part of the GFC. As in the case of GFCs, it also begins with the producers, the plants (directly or indirectly). However, the food energy passes from large organisms to small organisms in the parasitic chains. For instance, a tree which occupies the 1st trophic level provides shelter and food for many birds. These birds host many **ecto-parasites** and **endo-parasites**. Thus, unlike in the predator food chain, the path of the flow of energy includes fewer, large sized organisms in the lower trophic levels, and numerous, small sized organisms in the successive higher trophic levels.

III. Detritus Food Chain

The detritus food chain (**DFC**) begins with dead organic matter (such as leaf litter, bodies of dead organisms and faeces of animals). It is made up of **decomposers** which are **heterotrophic organisms**, mainly the 'fungi' and 'bacteria'. They meet their energy

and nutrient requirements by degrading dead organic matter or detritus. These are also known as **saprotrophs** (sapro: to decompose).

Decomposers secrete digestive enzymes that breakdown dead and waste materials (such as faeces) into simple absorbable substances. Some examples of detritus food chains are:

1. Detritus (formed from leaf litter)-Earthworms -Frogs -Snakes
2. Dead animals -Flies and maggots -Frogs -Snakes

In an aquatic ecosystem, GFC is the major ‘conduit’ for the energy flow. As against this, in a terrestrial ecosystem, a much larger fraction of energy flows through the detritus food chain than through the GFCs. Detritus food chain may be connected with the grazing food chain at some levels. Some of the organisms of DFC may form the prey of the **GFC** animals. For example, in the detritus food chain given above, the earthworms of the **DFC** may become the food of the birds of the **GFC**. It is to be understood that food chains are not ‘isolated’ always.

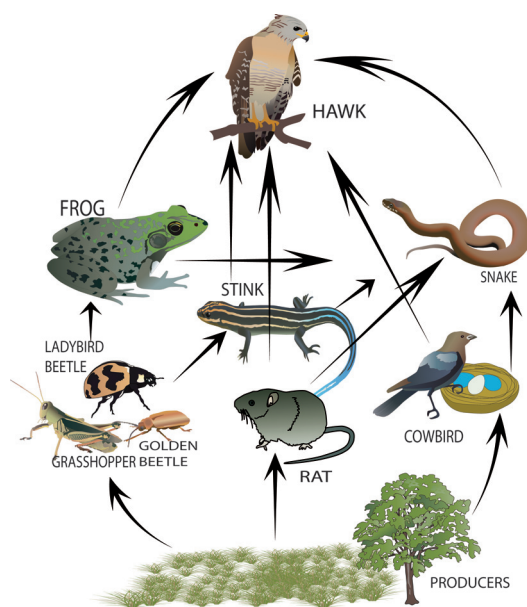


Fig. 8.15 A simple food web

8.5.2.1 Food Web

In a simple GFC the food energy flows through in **linear trophic levels**. However, in an ecosystem the feeding relationships are never that simple. The feeding interrelationships become more complex because of the omnivores present in the ecosystem. For instance man, birds and some other animals are ‘**omnivores**’, by their feeding habits. In such cases it is difficult to place an animal in a simple / single food chain as they have many interconnections with the animals of other food chains. These natural interconnections of food chains form a ‘net-work’ called ‘**food web**’.

8.5.3 Energy Flow

Except for the deep sea **hydro-thermal ecosystem**⁷, sun is the only source of energy for all ecosystems on Earth. Of the incident solar radiation less than 50 per cent of it is **photosynthetically active radiation** (PAR). We know that plants and photosynthetic bacteria (autotrophs), fix Sun’s radiant energy to synthesise food from simple inorganic materials. Plants capture only 2-10 percent of the PAR and this small amount of energy sustains the entire living world. So, it is very important to know how the solar energy captured by plants flows through different organisms of an ecosystem. All heterotrophs are dependent on the producers for their

⁷ Hydrothermal systems in the deep ocean, do not have light, and chemical energy supports life (chemoautotrophs), instead of sunlight.

food, either directly or indirectly. The law of conservation of energy is the **first law of thermodynamics**. It states that energy may transform from one form into another form, but it is neither created nor destroyed. The energy that reaches earth is balanced by the energy that leaves the surface of the earth as invisible heat radiation.

The energy transfers in an ecosystem are essential for sustaining life. Without energy transfers there could be no life and ecosystems. Living beings are the natural proliferations that depend on the continuous inflow of concentrated energy.

Further, ecosystems are not exempted from the Second Law of thermodynamics. It states that no process involving energy transformation will spontaneously occur unless there is degradation of energy. As per the second law of thermodynamics - the energy degraded is in the form of unavailable heat energy, and constitutes the **entropy** (energy lost or not available for work in a system). The organisms need a constant supply of energy to synthesize the molecules they require. The transfer of energy through a food chain is known as **energy flow**. A constant input of mostly solar energy is the basic requirement for any ecosystem to function. The important point to note is that the amount of energy available decreases at successive trophic levels. When an organism dies, it is converted to detritus or dead biomass that serves as a source of energy for the decomposers. Organisms at each trophic level depend on those at the lower trophic level, for their energy demands.

Each trophic level has a certain mass of living material at a particular time, and it is called the **standing crop**. The standing crop is measured as the **mass** of living organisms (**biomass**) or the number of organisms per unit area. The biomass of a species is expressed in terms of fresh or dry weight (dry weight is more accurate – because water contains no usable energy).

The 10 percent Law

The 10 percent law for the transfer of energy from one trophic level to the next was introduced by Lindeman (the Founder of the modern Ecosystem Ecology). According to this law, during the transfer of energy from one trophic level to the next, only about 10 percent of the energy is stored/ converted as body mass / biomass. The remaining is lost during the transfer or broken down in catabolic activities (Respiration). Lindeman's rule of trophic efficiency/ Gross ecological efficiency is one of the earliest and most widely used measures of ecological efficiency. For example, if the NPP (Net primary production) in a plant is 100 kJ, the organic substance converted into **body mass** of the herbivore which feeds on it is 10 kJ only. Similarly the **body mass** of the carnivore-I is 1 kJ only.

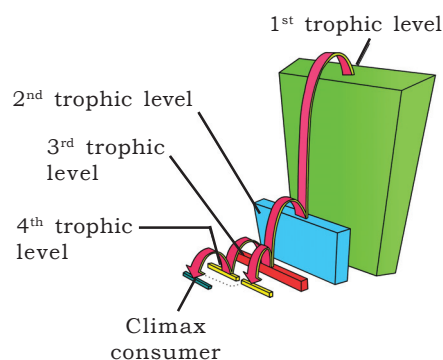


Fig. 8.16 Energy flow in an ecosystem

8.5.4 Productivity

The **rate of production** of biomass is called **productivity**. It can be divided into **primary** and **secondary** productivities.

- I. **Primary productivity** is defined as the amount of biomass or organic matter produced per unit area over a period of time by plants, during photosynthesis. It can be divided into gross primary productivity (GPP) and net primary productivity (NPP).
 - a) **Gross primary productivity** of an ecosystem is the rate of production of organic matter during photosynthesis. A considerable amount of GPP is utilized by plants for their catabolic process (respiration).
 - b) **Net primary productivity** Gross primary productivity minus respiratory loss (R), is the net primary productivity (NPP). On average about 20-25 percent of GPP is used for the catabolic (respiratory) activity.

GPP - R = NPP

The net primary productivity is the biomass available for the consumption of the heterotrophs (herbivores and decomposers).

- II. **Secondary productivity** is defined as the rate of formation of new organic matter by consumers.

8.5.5 Ecological Pyramids (Eltonian pyramids)

You must be familiar with the shape of a pyramid. The base of a pyramid is broad and it narrows down towards the apex. The trophic relationship is expressed in terms of numbers; biomass or energy arranged one on the top of its lower trophic level, resulting in a pyramidal shape. It is a graphical representation of the trophic structure and function of an ecosystem. The base of each pyramid represents the **producers** or the first trophic level, while the apex represents the tertiary or top level / top order consumer. The three types of ecological pyramids that are usually studied are (a) **pyramid of number**; (b) **pyramid of biomass** and (c) **pyramid of energy**. These pyramids were first represented by **Elton**, hence the name **ELTONIAN pyramids/ Ecological pyramids**.

Any calculations of energy content, biomass, or numbers has to include all organisms at that trophic level. No generalizations we make will be true if we take only a few individuals of any trophic level into account. In most ecosystems, all the pyramids- of numbers, energy and biomass are **upright** i.e., producers are more in number and biomass than the herbivores, and herbivores are more in number and biomass than the carnivores. Also energy (available) at a lower trophic level is always more than that at a higher level.

There are exceptions to this generalization. In the case of a parasitic food chain, the pyramid of numbers is **inverted**. A large tree (single producer) may support many

herbivores like squirrels, and fruit eating birds. On these herbivores many ectoparasites such as ticks, mites and lice (secondary consumers) may live. These secondary consumers may support many more top level consumers and also the hyperparasites. Thus in each trophic level from the bottom to the top, the numbers of organisms increase, and form an 'inverted pyramid' of **numbers**.

The pyramid of **biomass** in sea is also generally inverted because the biomass of fishes far exceeds that of phytoplankton (Ref: NCERT Text Book).

Pyramid of **energy** is always **upright**, and can never be inverted, because when energy flows from a particular trophic level to the next higher trophic level, some energy is always lost as heat (thus at every step). Each bar in the energy pyramid indicates the amount of energy present at each trophic level in a given time or annually per unit area.

Limitations of Ecological Pyramids

However, there are certain limitations of ecological pyramids, such as –

- it does not take into account the same species belonging to two or more trophic levels,
- It assumes a simple food chain, something that almost never exists in nature,
- it does not accommodate a food web, iv) moreover, saprophytes are not given any place in ecological pyramids even though they play a vital role in the ecosystem.

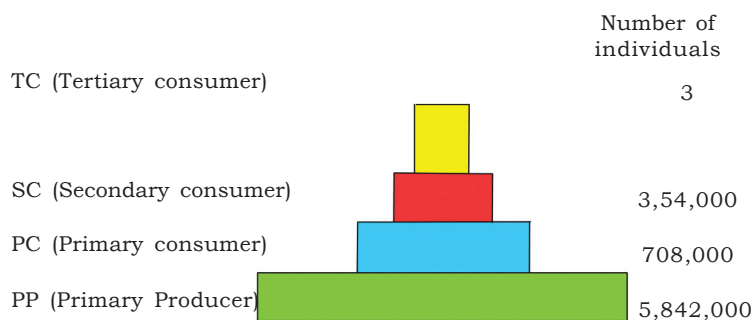


Fig. 8.17 (a) Pyramid of Numbers in a grassland ecosystem. NOTE: Only 3 top-carnivores are supported in an ecosystem based on production of nearly 6 million plants. Also NOTE that the pyramid of numbers of a parasitic food chain is inverted.

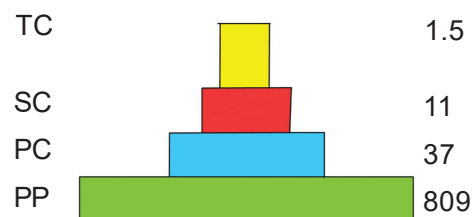


Fig. 8.17 (b) Pyramid of biomass shows a sharp decrease in biomass at the higher trophic levels. NOTE: Pyramid of biomass is upright normally.

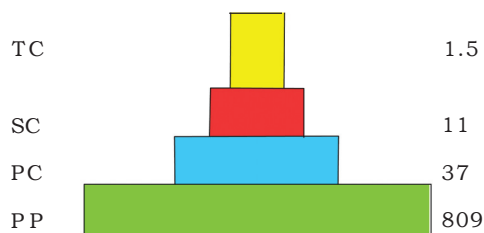


Fig. 8.17 (c) An ideal pyramid of energy which is an always upright pyramid. NOTE: primary producers convert only 0.5% of the energy absorbed from the sunlight available to them are into GPP

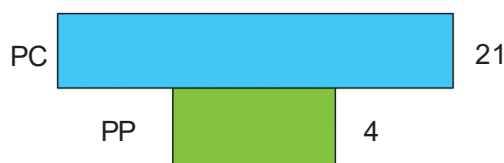


Fig. 8.17 (d) Inverted pyramid of biomass-small standing crop of phytoplankton supports large standing crop of zooplankton (at certain times) NOTE: Algal blooms are followed by their death due to short life span depleting their numbers (temporarily).